

INSTITUTE OF ACTUARIES OF INDIA

Subject CP1 – Actuarial Practice (Paper A)

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

a) There are two main approaches to financing benefits – i.e., unfunded and funded

By **unfunded** approach the sponsor of the scheme generally pays the benefits as and when they fall due and by finding the money at the point of payment.

The unfunded approach is called pay-as-you-go.

In the **funded** approach, the sponsor sets aside the money in advance to fund the benefit costs when the benefits fall due.

[2]

b) Types of funded approach:

- Lump sum in advance – Lumpsum is paid in advance and this is believed to be sufficient to provide the benefits when it falls due. E.g. purchase of annuities
- Terminal funding - a payment is made whenever a benefit starts to be paid. The payment is a capital sum, designed to be sufficient to provide all the future payments of the benefit.
- Regular contributions - Funds are gradually built up to a level expected to be sufficient to meet the cost of the benefit, over the period between the promise being made and the benefit first becoming payable
- Just-in-time funding - payment is made (as the name suggests) at the last possible moment. Payment is triggered by an external event (not a benefit event). E.g. Insolvency of employer
- Smoothed PAYG - Funds that are set up to smooth the costs under a pay-as-you-go approach to allow for the effects of timing differences between contributions and benefits

[5]

[7 Marks]**Solution 2:**

a) The key liability profiles of a general insurance company differ from that of a life insurance company in several ways:

1. **Nature of Risks:** General insurance companies primarily deal with short-term risks such as property damage and accidents. In contrast, life insurance companies deal with long-term risks related to mortality, longevity, and morbidity.
2. **Policy Duration:** General insurance policies are typically short-term, lasting for a year or less, whereas life insurance policies are long-term, often lasting for several years or even decades.
3. **Frequency of Claims:** General insurance policies often have a higher frequency of claims due to the nature of the risks they cover, such as accidents and property damage. Life insurance policies, especially pure life policies, may have lower claim frequencies but higher claim amounts.
4. **Reserving Requirements:** General insurance companies need to set aside reserves to cover future claims, but these reserves are generally shorter-term (except for Third party liability claims) compared to life insurance policies.

[4]

b) As a result of these differences, the investment strategy and portfolio of a general insurance company differ from that of a life insurance company in the following ways:

1. **Liquidity Needs:** General insurance companies may need to maintain more liquid investment portfolios to meet short-term claim obligations.
In contrast, life insurance companies may have more flexibility to invest in less liquid assets due to the longer-term nature of their liabilities.
2. **Risk Profile:** General insurance companies may focus more on generating stable and predictable returns to meet short-term obligations and manage underwriting risks. Life insurance companies may have a longer investment horizon, allowing them to take on more risk to achieve higher long-term returns.
3. **Asset Allocation:** General insurance companies may allocate a larger portion of their portfolios to fixed-income securities and cash equivalents to ensure liquidity.
Life insurance companies may afford higher allocation to equities and alternative investments, some of which may be less liquid, to enhance returns over the long term.
4. **Duration Matching:** General insurance companies may not need to match the duration of their assets with the duration of their liabilities as closely as life insurance companies, which often use duration matching strategies to manage interest rate risk.

[4]

c) Points to highlight regarding investing cash assets in gold:

- The factors that contributed to the rise in gold prices, such as
 - o inflation concerns,
 - o geopolitical tensions, and
 - o currency fluctuations
 may not continue in the future which can lead to lower than average or even negative returns
- Gold prices can be more volatile than money markets which can lead to losses at the time of sale
- Gold is generally priced in US dollars. Any fluctuations in the currency could also make the prices more volatile.
- Opportunity cost: Past performance of the asset need not replicate in the future. There could be other asset classes, e.g. equity that could yield better returns. The company may miss out on this if it solely invested based on the last 1 year's performance of gold.
- Physical gold may not be as liquid and there could be storage risks and insurance and transaction costs of dealing in physical gold.
- Investing 75% of the cash assets in physical gold may also hinder the company's ability to meet its liabilities especially if it plans to write short tailed lines of general insurance business such as personal accident, motor own damage and household contents insurance
- Alternatives such as ETFs or gold bonds can be considered which are more easily tradable and does not involve overheads.
- Diversification needs to be considered. If the company plans to have enough other investments in short term liquid assets, then the remaining cash assets could be invested in gold
- Thresholds and investment strategies can be defined which could allocate assets to or away from gold instead of just investing the current 75% of cash in gold.
- Gold is viewed as a hedge against inflation. If the company also plans to underwrite long tailed liability lines of business, then some funds could be allocated to gold.
- Current economic environment: interest rates and gold prices are negatively correlated with each other as investors. If the company expects interest rates to fall in the future, then some funds could be allocated to gold.
- Need to analyse correlation with returns of the equity market and other broader macroeconomic factors before investing a large sum in gold

[6]

[14 Marks]

Solution 3:**a) Medical**

- Medical History: Evaluation of the applicant's medical history, including past illnesses, surgeries, and chronic conditions
- Family medical history
- Current health status and whether any illnesses are already existing

Lifestyle

- Since the company is selling term insurance it is at a risk of excessive deaths than has been priced for hence it is essential for the company to understand the quality of lives being underwritten
- Smoker status – Since this reduces lifespan it is important to determine appropriate loadings for smoker
- From a risk management perspective it is important to maintain a balance of smoker lives in the portfolio
- Occupation of the applicant will help determine whether the applicant is exposed to riskier occupations such as hazardous waste handling or mining
- Hence, appropriate underwriting conditions or loadings to the premium can be applied.

Financial

- Income verification – this is needed to ensure that the coverage being opted for is in line with the income of the applicant ensuring that the probability of fraudulent claims are reduced.
- Existing Insurance Coverage: Checking the applicant's existing life insurance coverage to determine if they are adequately insured and to prevent over-insurance, which can indicate potential for moral hazard.
- Debt and Liabilities: Reviewing the applicant's financial liabilities and debts to assess their financial stability and the risk of default on premium payments.

[4]

b) Diversification

- Across the population to which the product is being sold for example:
 - o Age group
 - o Gender
 - o Smoker status
- Sales channel – use variety of modes of selling policies such as bancassurance, digital, agents, brokers
- Across region
- Use reciprocal quota share to get diversification across other products
- Start other product lines
- Invest in a diverse range of asset classes to spread the risk and optimise the return

Underwriting

- Design the proposal form in a way that elicits truthful responses from the customers
- Identify risks that need special terms or reduced limits
- Monitor actual vs expected and ensure that deviations are minimised
- Business mix monitoring
- Rate adequacy monitoring

Claims control

- Have frameworks and processes in place to identify / flag high risk of fraud claims
- work with government agencies to ensure only genuine claims are paid out
- Claims process to ensure customer grievances are minimised

Management controls

- Ensure correct data has been collected so that at the time of claiming the risk of repudiations is minimised
- Regular audits of processes
- Feedback of reserving into the rating processes

Other measures

- Excess of loss reinsurance
- Asset liability matching
- Customer retention

[8]

[12 Marks]

Solution 4:

- a) This incident is likely to have significant implications for various types of insurance classes, including:
- **Marine Hull Insurance:** This type of insurance covers physical damage to the ship itself. In this case, the ship that collided with the bridge would be covered under marine hull insurance. The impact of the incident on marine hull insurance would include:
 - o Payout for the repair or replacement of the damaged ship.
 - o Potential salvage costs for removing the ship from the accident site.
 - o Any additional costs incurred for towage or other necessary services.
 - o Potential loss of income if the ship is unable to operate while repairs are being carried out.
 - **Marine Liability Insurance:** This type of insurance covers liability for damage caused by the ship to third parties or their property. The impact of the incident on marine liability insurance would include:
 - o Payouts for damage to the bridge and any other property caused by the ship.
 - o Compensation for any injuries or fatalities resulting from the accident.
 - o Legal costs associated with defending against liability claims.
 - o Potential environmental cleanup costs if the accident leads to pollution.
 - **Property Insurance:** The owners of the bridge and any other property damaged in the accident may have property insurance to cover the costs of repairs or replacement. Insurers providing property insurance would be impacted by the payouts for these claims.
 - o Removal of debris
 - **Business interruption**
 - o Losses resulting from the closure of the bridge
 - o Costs involved in opening a corridor to let the other ships pass while the bridge is rebuilt

Legal aspects

- Determining the cause of the accident will be difficult
- Complex litigation process as different countries maybe involved with different laws leading to jurisdiction issues
- Multiple parties may be involved hence settlement is difficult
- Long litigation process
- Difficult to ascertain whether claims being reported are they directly related to the incident
- Moreover, there may not have been events of this scale in the recent past that could act as landmark cases for judgements
- Fraudulent claims may be reported
- The ship owner, ship manger may be based in different countries or continents.
- Due to the involvement of multiple parties, settlement is difficult as it cannot be done with a single party

- Who and which class of insurance pays for damage as a result of the accident will further add to the complication.

Reinsurance

- Insurers will look to recoup some of the losses from reinsurance.
- The reinsurance structures and limits for different classes of insurance will be different.
- Moreover, such large risks will be shared between multiple reinsurers
- May involve reinsures across various jurisdictions

Government

- o may need to pay for a temporary opening for continuous trade and to reduce the impact of losses
- o May also need to pay for the construction of the new bridge for both political and financial purposes.

[14]

b) Impact on marine insurance and reinsurance rates

Marine insurance rates:

- The immediate impact of the incident will lead to increase in marine insurance rates.
- However, this may also lead to more stringent terms and conditions from reinsurers on this class of business.
- There may not be long term impacts on the rates if the losses to the industry are lesser than expected and the government foots the bill of reconstruction.
- The total loss from the bridge collapse may be lesser than that from natural catastrophes. Thus the industry may be able to absorb these losses without much financial strain on the balance sheet

Impact on reinsurance rates:

- Marine insurance is largely reinsured hence reinsurers will drive the rates post-accident
- The short-term rates may increase as reinsurers will be unaware of the extent of loss
- As the loss is quantified and the liabilities are assigned the rates may be revised in the long term to a reasonable level
- Thus in the long term the rates may not change significantly as reinsurers may have created reserves for this type of accident over a number of years

[3]

[17 Marks]

Solution 5:

a) Regulatory actions to promote innovation:

The insurance regulator in the country of Utopia can promote innovation in the insurance industry through the following measures:

Regulatory Sandbox: Implementing a regulatory sandbox where insurers can test innovative products, services, and business models in a controlled environment with exceptions to the current regulations.

Innovation Challenges: Organizing innovation challenges or competitions to encourage insurers to develop creative solutions to address specific challenges or opportunities in the market.

Partnerships with Insurtechs: Encouraging partnerships between traditional insurers and insurtech startups to leverage technology and data analytics for developing new products and enhancing customer experiences.

Customer-Centric Products: Encouraging insurers to develop customer-centric products that are tailored to meet the specific needs and preferences of different customer segments.

Flexible Regulatory Framework: Creating a flexible regulatory framework that allows for innovation while ensuring consumer protection and financial stability.

Education and Awareness: Educating consumers about the benefits of insurance and the importance of risk management, which can help increase insurance penetration.

Deregulation:

- Reducing reporting requirements
- Easing launch of new products by introducing Use and File
- Detariffication/ price decontrols

[5]

b) Impact of decontrol of pricing:

- Segments that were unprofitable to the entire industry may see rate increases or differentiated product offering leading to improved profitability from such segments
- New products may be designed that cater to emerging technologies such as Electric vehicles, Hybrid and Hydrogen fuel vehicles.
- Decontrol may also see improved service levels from insurers – as customers can be given options for the method of claiming – reimbursement, fixed/parametric
- DigitalCo may be able to have a price differentiation given its distribution strategy of digital only
- It may have more cost efficiencies which may be passed on to the customers with more fine pricing
- Specialised fronting companies may come up with tailored products and better understanding of the risk
- Education may be needed from the regulator / industry bodies to increase customers awareness
- DigitalCo for sometime may continue to sell the old product as its customers are aware of the old product.
- DigitalCo may have better reinsurance rates which could help it compete better in the market with refined pricing
- There may be other competing or innovative insurance companies to DigitalCo that would reduce prices to gain market share. Thus, rate competition may lead to deterioration in loss ratios – ultimately impacting solvency.
- As motor is a large line of business for most insurers this impact maybe significant.
- DigitalCo may face difficulty selling new policies as customers may be used to assisted selling...
- ...hence they may not understand the new products –
Until now customers were protected by the standard / tariff policy wordings. Now, grievances may increase due to ambiguity in products/policy wordings and as a result increase in operational costs for the insurers
- Regulatory challenges: Monitoring needed from regulator to ensure fair and transparent practices
- On the other hand, it may face more competition from more tech savvy companies or traditional companies which may drive price more aggressively in the market

[8]

c) Investment in the company

- The insurance market maybe growing and DigitalCo. maybe the leading player for this distribution segment in the insurance market.

- High growth potential for retail products such as Motor and Health on the digital channel
- Diversification: Investing in DigitalCo. may provide the investor with diversification benefits, especially if they have a portfolio of investments across different industries and asset classes.
- As the company is new, they may be spending more to build the brand through spends on social media ads and marketing campaigns
- The combined ratio may be distorted by higher initial costs of technology set-up, licenses
- Renewal rate is better on digital channels due to access to customers and the cost of renewing policies on digital channel is low **(1 mark)**
- Conversion rates for DigitalCo may be better than the average / market
- They maybe catering to a unique specific segment only (eg. Radio cab/High age groups/Electric vehicles) which have good growth potential but ...
- ...may not be preferred by insurers currently
- To address such segments DigitalCo may have a new differentiated product offering with experts to run this segment sustainably.

[5]

[18 Marks]**Solution 6:****a) Key areas considered by the hospital chains**

- Scope of the health insurance market
- Brand Reputation and Trust: The hospital may have a good brand name within health industry in the country and the customers would trust buying health insurance from such a company
- Readily available health and treatments inhouse data with hospital chains
 - o Cost of treatments
 - o Most prevalent diseases
 - o Trends in new diseases
- Synergies with existing operations
 - o Expertise in preventive health and wellness programs that can help reduce claims frequency/severity
 - o Existing base of staff – doctors and nurses that can be trained to move as UW and claims experts on the health insurance business.
 - o Synergies in backend operations – using the group’s existing administrative structure to bring synergies
- Efficiency in servicing claims: Keep claim servicing cost and premiums within the group.
- Flexibility to design product customised to needs of insureds
- Existing customer base that can be pitched customised products
- Strong financial controls from management
- The hospital chain may not have any prior experience in insurance
- Delay in regulatory approvals – which may derail the business plan diluting the overall returns
- Difficulty in attracting insurance talent/business acumen
- Talent availability
- Capital availability – if sales are not as expected, then shareholders should have enough capital to fund for that period.
- As the hospital chain does not have expertise in the insurance sector and dealing with long tailed liability risks its risk appetite to absorb losses may be low

- They may not know to effectively manage anti-selection/ fraud claims
- Strong management control set up is needed

[9]

b) Policy and claims data fields

Policy:

- Date of issue
- Individual / Family floater
- Age
- Gender
- Pre-existing conditions
- Sum insured
- Address/Location
- Deductible

Claims:

- Date of loss
- Number of days in hospital
- Claim amount
- Treatment performed

[3]

c) Issues with industry data

- Different companies may have different products and coverages which may not align with the product that the new company may want to offer to its customers
- Detail – level of detail in the industry level data may not be enough rendering the data not useful for the purposes of pricing
- Risk factor coding may be different across companies.
- Out of date – Regulation changes e.g. change in waiting period, emerging trends in diseases may not be captured by industry data if the collection is not frequent.
- Differences in sales method, socio economic groupings, storage of data
- Not everyone contributes and distorted by data of large player being missing
- Quality of data – depends on contributors. Different companies have different maturities of data systems and the quality of data may not be consistent leading to no beneficial gain from the data

Overcoming these challenges:

- Hire experts from the insurance industry / competitors that have a view on the probability of occurrence (frequency & severity) and cost of treatments that the products of the company aims to cover.
- Validate these with the industry data that is available.
- Taking external help - a consultant can be employed to provide views
- Reinsurer's views
- Collaboration from parent company in case of joint venture / partner who can provide the expertise.

[4]

[16 Marks]

Solution 7:

a) Long term bond investment by a General insurer

- Large casualty/liability book – Motor, Product liability. Match income from long term bonds to meet the payment needs of liability businesses.

- Better returns on long term bonds
- Free capital and bond prices attractive now
- Liquidity management: While long-term bonds have a fixed maturity date, they can still provide liquidity if needed through secondary markets, albeit with potential price fluctuations.
- Diversification: Including long-term bonds in the investment portfolio helps diversify risk, reducing exposure to any single asset class or market segment.
- Credit quality: Long-term bonds issued by governments or highly rated corporations are typically considered safe investments, offering insurers a way to preserve capital while generating income.

[2]

b) Quota share helping an insurer

- How it improves capital – part of liability taken to reinsurer at a lower rate
- Spreads risk and enables insurers to write larger portfolio of risks
- Improves solvency margin – as net written premium reduce and the Sum at Risk whilst free assets remain constant
- Commission income from quota share may help with cashflow and financial management, thus improving RSM because of reinsurance credit.
- Volatility management
- Takes away the claims liability
- Shares large claims

[3]

c) Reinsurer's statement evaluation

- Reinsurer's margin would be less than the cost of interest on the subordinated debt. Cost of interest is 10% against paying 5% margin to the reinsurer which comes with additional benefits.
- The company is not looking for cash relief but a capital management solution where reinsurance could help. The commission from the reinsurer will also help with part cash relief
- Reinsurer can help with volatility management via XoL arrangements
- Time involved in raising debt vs securing a QS – QS negotiation maybe faster as insurer is looking for short term capital relief
- Cost of raising debt vs securing quota share – cost of raising debt will be higher. Merchant banking costs.
- Already have a strong and longstanding partnership with a reinsurer and can leverage that to secure good terms
- Additional benefits and market insights as a result of working with reinsurer
- Reinsurer may provide analytics services and an outside-in view of the company's business
- Regulatory approvals – may be easier to get approvals for RI rather than debt raising and also debt raising may involve multiple regulatory bodies such as capital market regulator.

[4]

d) Profit commission

Commission paid by a reinsurer to a cedant under a proportional reinsurance treaty that is dependent upon the profitability of the total business ceded during each accounting period. It is also used in other arrangements, such as commission contingent on claims experience.

[2]

e) Capital management tools

Financial reinsurance: For example, a contingent loan where the repayments of the loan are contingent on a certain event such as, the insurance company making profits in a certain line of business or portfolio.

- Because the insurance company has no liability to repay the loan unless these profits emerge, it does not have to make any provision for these future payments on a statutory basis. Therefore, the insurance company improves its statutory solvency position.
- As the company will repay the loan over time, the improvement is short term and meets the needs of the insurer

Securitisation : involves converting an illiquid asset into tradable instruments. The primary motivations are often to achieve regulatory or accounting “off balance sheet” treatment. - \

- Transactions can be structured with an a smaller element of risk transfer associated with the value of the asset.
- This will result in any potential loss in value of the asset being capped.
- The insurer can securitize the future profits/premiums to get up-front capital

Contingent capital: can be a cost-effective method of protecting the capital base of an insurance company. Under such an arrangement capital would be provided as it was required following a deterioration of experience (i.e. it is provided when it is needed). Since the company doesn't need immediate cash relief, this could be used to fund the cash when needed.

Derivatives: Structured derivatives that protect the insurer against further losses in the bond portfolio could be used to protect asset values

Traditional forms of capital – equity and debt – Though the insurer doesn't seem to need immediate cash relief, it may raise equity to provide for the long term. There maybe capital available now and the insurer may want to capitalise on it rather than wait till there is a large loss which may force the insurer to sell its bonds at losses.

Internal sources of funding: Due to the increase in interest rates, if the insurer is allowed to discount liabilities in the accounting standards, this could be discounted at the higher interest rates leading to lower liability values.

[5]

[16 Marks]
